Brake Control Valve – General Data

1. General

The brake control valves are connected through a system of push rods, levers, and torsion rods to the brake pedals on each rudder pedal. A turnbuckle is installed in each push rod to increase pressure adjustment for the brake control valve. The valves are located on the flight compartment floor, and are in effect variable pressure reducers which meter fluid from the hydraulic system to the brake pressure lines in proportion to brake pedal movement. Each valve consists of a cast aluminum alloy body containing a lever-operated floating piston and a spring-loaded poppet valve. The valves are connected into the system through three ports, inlet pressure brake pressure, and return (refer to 29-10-00, Hydraulic Power Package (Mod 6/2039) – General Data).

2. Removal/Installation

Refer to Figure 1 for details.

A. Remove Brake Control Valve

1) Discharge brake system accumulator hydraulic pressure by pumping brake pedals.

2) Remove access panels in flight compartment.

3) At brake control valve, disconnect brake system indicator hydraulic line (right-hand valve only); disconnect inlet pressure, brake pressure and return hydraulic lines. Cap open connections and lines.

4) Remove cotter pin, nut, washer and bolt attaching brake control linkage levers to mounting bracket. Withdraw spring torsion rod (short) from brake control valve lever.

Note When removing right-hand brake control valve remove bolt at right-hand linkage levers and mounting bracket. For left-hand valve, remove bolt at left-hand linkage levers and mounting bracket.

5) Remove four bolts securing brake control valve to flight compartment floor, and move valve outboard to disengage valve lever from spring torsion rod (long).

B. Install Brake Control Valve

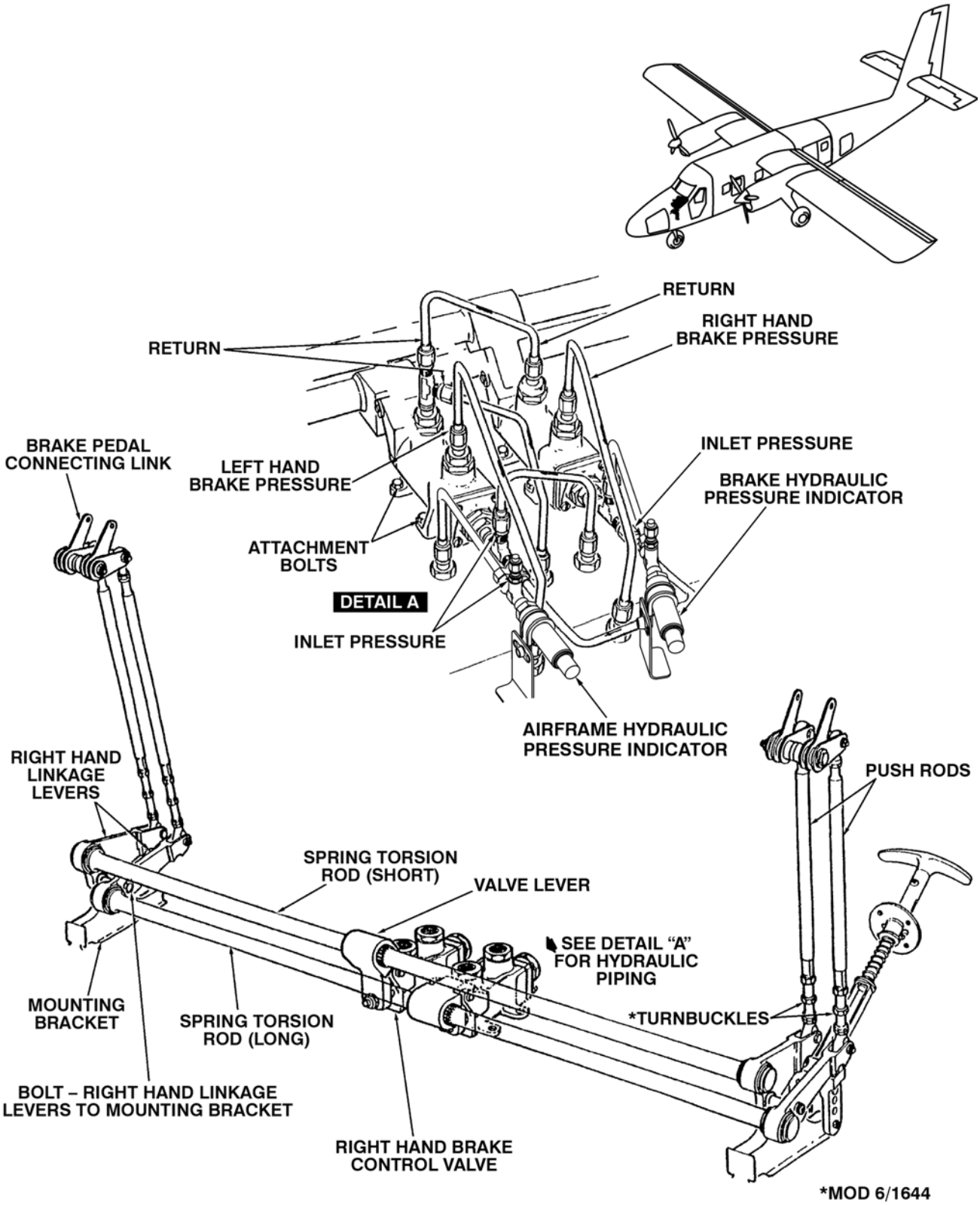
1) Engage valve lever on spring torsion rod (long), and position brake control valve on flight compartment floor. Install four bolts securing valve to floor.

2) Engage spring torsion rod (short) with valve lever. Position control linkage levers at mounting bracket and install bolt, washer, nut and cotter pin.

3) At brake control valve, connect brake system indicator hydraulic line (right-hand valve only); connect inlet pressure, brake pressure and return hydraulic lines.

4) Bleed brake system and function test wheel brakes. (Refer to 32-40-00, Wheels and Brakes (Mod 6/2002) – General Data).

5) Install access panels in flight compartment.



Brake Control Valve Installation
Figure 1